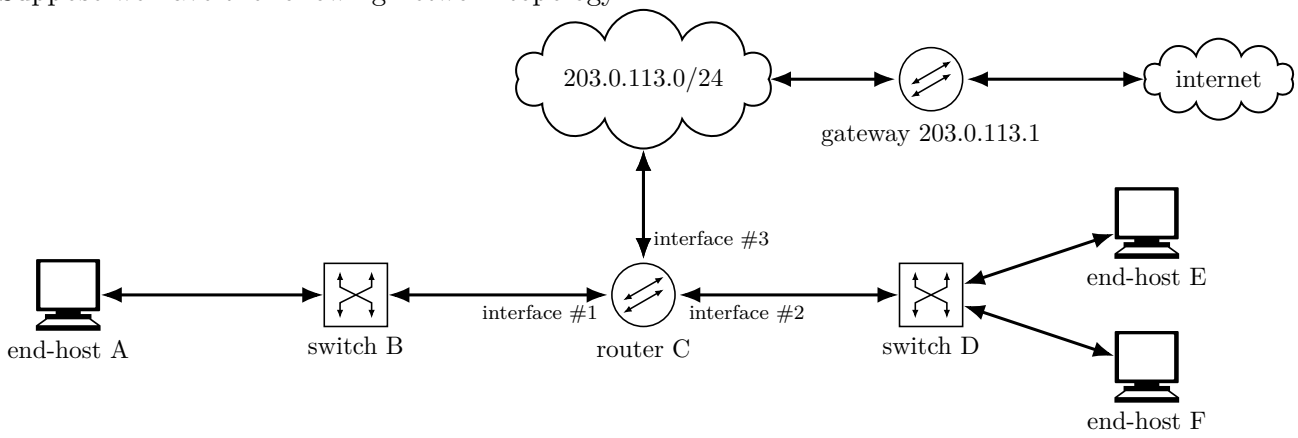
Name: _____

Write your name and computing ID above. Write your computing ID at the top of each page in case pages get separated. Sign the honor pledge below.

On my honor as a student I have neither given nor received aid on this exam.

_____

1. Suppose we have the following network topology:



(or described without a picture: end-host A and router interface 1 are connected to switch B; end-host E and F and router interface 2 are connected to switch; and router interface 3 is connected to the internet via the 203.0.113.0/24 network and the gateway 203.0.113.1).

On these networks:

- the end-host and router interfaces are assigned IP and MAC addresses as follows:

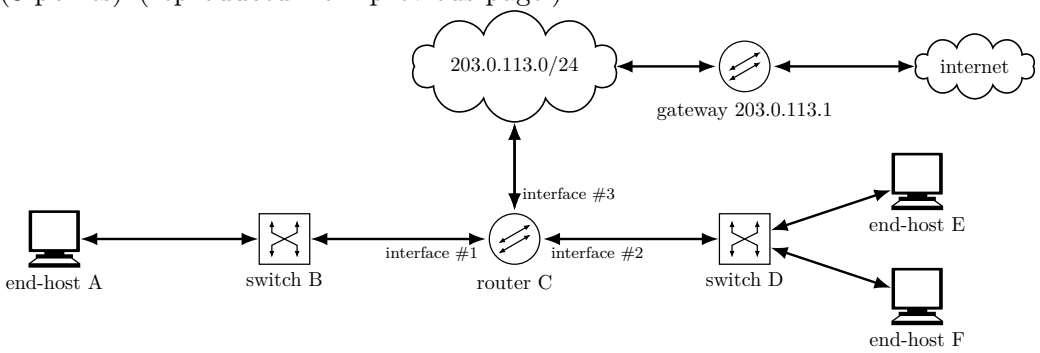
node/interface	IP	MAC
end-host A	`198.51.100.2`	`11:22:33:44:aa:bb`
end-host E	`192.0.2.2`	`11:22:33:44:aa:cc`
end-host F	`192.0.2.3`	`11:22:33:44:aa:dd`
router C interface 1	`198.51.100.1`	`11:22:33:44:11:22`
router C interface 2	`192.0.2.1`	`11:22:33:44:11:23`
router C interface 3	`203.0.113.5`	`11:22:33:44:11:24`

- the router uses a gateway of 203.0.113.1 to contact hosts on the internet not in the 203.0.113.0/24 network

(a) (8 points) Complete a likely routing table for router C. Use CIDR notation (e.g. `1.2.0.0/16`) for all address ranges. Use routes that cover a range of addresses instead of individual addresses whenever practical. You may not need all lines.

destination IPs	gateway (if applicable)	interface #
198.5.100.0/24 or similar	N/A	1
192.0.2.0/24 or similar	N/A	2
203.0.113.0/24	N/A	3
0/0 or default	203.0.113.1	3
—	—	—
—	—	—
—	—	—

(b) (8 points) (reproduced from previous page:)



node/interface	IP	MAC
end-host A	`198.51.100.2`	`11:22:33:44:aa:bb`
end-host E	`192.0.2.2`	`11:22:33:44:aa:cc`
end-host F	`192.0.2.3`	`11:22:33:44:aa:dd`
router C interface 1	`198.51.100.1`	`11:22:33:44:11:22`
router C interface 2	`192.0.2.1`	`11:22:33:44:11:23`
router C interface 3	`203.0.113.5`	`11:22:33:44:11:24`

Suppose end-host A port 5555 sends a UDP packet to end-host E port 6666.

Fill in the table of the source and destination MAC addresses, IP addresses, and port numbers that will be present in those frames. For MAC addresses, please write only the last 16 bits (4 hexadecimal digits).

If multiple frames are sent with identical attributes, you may omit duplicate frames.

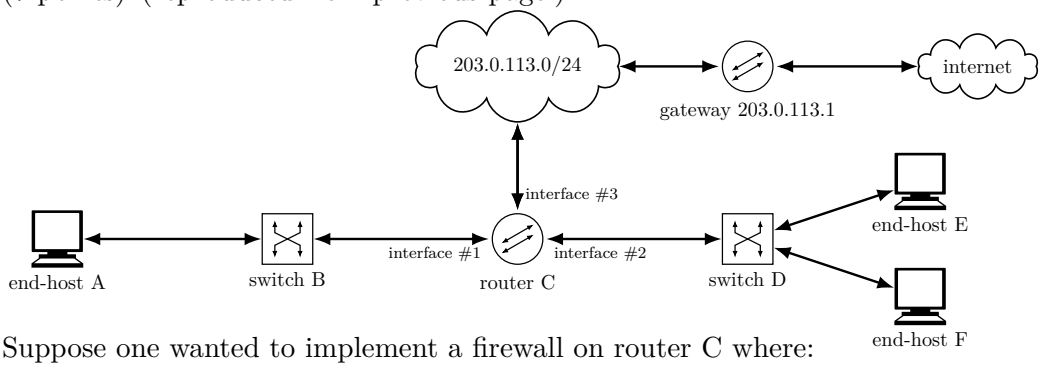
Assume that any ARP (address resolution protocol), DHCP, and related setup has already completed.

If you need to know an value that is not specified above, indicate something like 'X's MAC address'. If some field won't be present in the frame, write 'N/A'.

You may not need all lines.

frame #		source	destination
1	last 16 bits of MAC	aa:bb	11:22
	IP address	10.1.0.2	10.2.0.2
	port numbers	5555	6666
2	last 16 bits of MAC	11:23	aa:cc
	IP address	10.1.0.2	10.2.0.2
	port numbers	5555	6666
3	last 16 bits of MAC		
	IP address		
	port numbers		
4	last 16 bits of MAC		
	IP address		
	port numbers		

(c) (7 points) (reproduced from previous page:)



Suppose one wanted to implement a firewall on router C where:

- the firewall allows
  - end-host E to connect to TCP port 9999 on end-host A and then send and receive data on that connection,
  - end-host A to connect to any TCP port on end-host E or F and then send and receive data on that connection
  - end-host E to send UDP packets to UDP port 4444 on end-host A and receive responses from end-host A UDP port 4444, and
  - any 'control' packets like acknowledgments and ICMP messages related to the above
- and the firewall drops all other packets between end-host E and A.

To implement this policy, router C would need to _____. **Select all that apply.**

- ☐ change the TCP port number of some packets sent by end-host E to end-host A
- ■ examine the TCP flags of some TCP packets sent by end-host E
- ☐ examine the sequence numbers of some TCP packets sent by end-host E
- ☐ limit the maximum size of packets sent by end-host A smaller than it would be otherwise
- ■ track some information about each active TCP connections between end-host E and A
- ☐ track some information about each of the recent UDP messages from end-host E to A
- ☐ adjust the configuration of switch B and/or switch D *was switch C on original exam*

2. Suppose three flows A, B and C with equal-sized packets go through a shared link, and the router that feeds that link uses weighted fair queuing with equal weights for each flow (and drops the packet with highest virtual time (scheduled to transmit last) when the queue is full). The link can handle 1000 packets/second. Flow A attempts to send 300 packets/second; flow B attempts to send 300 packets/second; and flow C attempts to send 500 packets/second.

(a) (2 points) What is the approximate throughput (after accounting for lost/delayed packets) of flow A? You may leave your answer as an unsimplified arithmetic expression.

Solution: 300 packets/second

(b) (3 points) What is the approximate throughput (after accounting for lost/delayed packet) of flow C? You may leave your answer as an unsimplified arithmetic expression

Solution: 400 packets/second

3. Suppose there is an active TCP connection between some host A and some host B and packets with the following attributes of packets sent from A to B:

#	from	to	sequence number	acknowledgment number	data size	dropped?
1	A	B	500	2310	1000	no
2	A	B	1500	3310	0	no
3	A	B	1500	4310	0	no
4	A	B	1500	4310	1000	no
5	A	B	2500	4310	1000	no
6	A	B	3500	4310	1000	yes
7	A	B	4500	4310	1000	no
8	A	B	5500	4310	500	no

Assume that only packets labeled as 'dropped' are lost for any reason.

Recall that in TCP, the sequence number represents the **byte number** of the first data byte in that packet (so a 10 byte packet with sequence number 10 contains byte numbers 10 through 19, inclusive), and an acknowledgment number contains the byte number of the next expected sequence number (so acknowledgment number 20 acknowledges bytes up to and including 19).

(a) (4 points) Suppose **B's** sending window size was initially 2000 and it was not in 'slow start'. Assuming B most recently sent two 1000-byte data packets and was using the standard loss-based TCP congestion control with additive increase (by around 1 packet per round-trip-time) and multiplicative decrease (by a factor of two on loss or timeout), we would expect **B's** sending window size to _____ as result receiving A's messages above.

- ○ increase to around 2500
- ■ increase to around 3000
- ○ increase to around 4000
- ○ decrease to around 1000
- ○ something else, _____

(b) (4 points) Suppose A sends the above packets without receiving packets from B between sending the first and last packets above. What is the **minimum** sending window size (in bytes) A could have?

5500

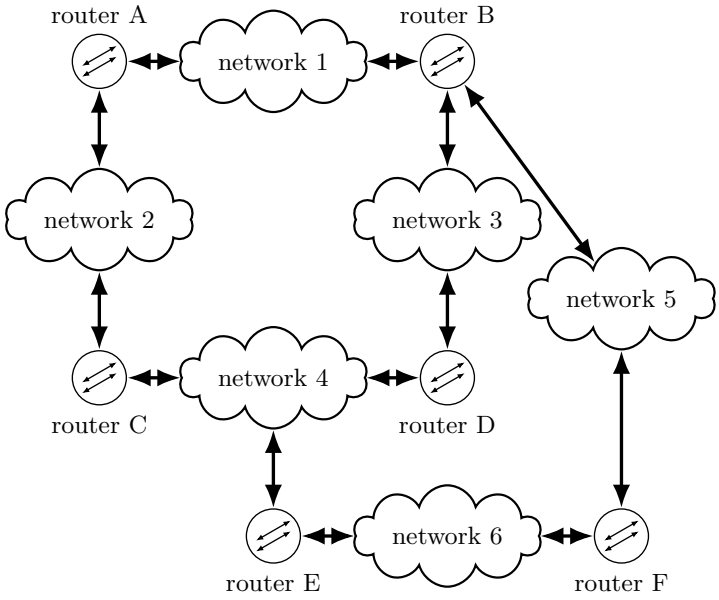
(c) (4 points) Packets # 2 and #3 were sent by A in response to packets B sent. What could have been attributes of those packets?

from	to	sequence number	acknowledgment number	data size
B	A	less than 3310	500	???
B	A	3310	500	1000

(d) (8 points) Suppose A receives 5 acknowledgment packets from B, one for each of the data packets B receives . These will have acknowledgment numbers 1500, 2500, 3500, 3500, and 3500. **Fill in the blanks and circle** an option where indicated below.

As a result of receiving these packets and any timeouts related to A's data packets expiring, A will (circle one) increase / decrease / not change its window size. The next data packet A will send will have sequence number 3500 . After that, the following data packet A would send will most likely have sequence number 6000 . *assuming we have new data to send and no major problems receiving the ACK for the retranmitted packet before any timeouts*

4. Consider the following network topology (shown in picture or with equivalent text description):



router A is connected to and	network 1 network 2
router B is connected to and and	network 1 network 3 network 5
router C is connected to and	network 2 network 4
router D is connected to and	network 3 network 4
router E is connected to and	network 4 network 6
router F is connected to and	network 5 network 6

C and D swapped in table above on exam as printed

(a) Suppose the routes were setup via a distance vector protocol, where each router is considered to have a cost of 1 to reach the networks it is connected to and any router attached to any of those networks. Assume the 'split horizon with poison inverse' approach is in use, where distance vectors advertized to neighbors have a cost of infinity for routes that go through that neighbor. (For example, if D's route to network 6 uses router E, it will advertise a distance vector with cost ∞ for network 6 to router E, but one with a finite cost to router A.)

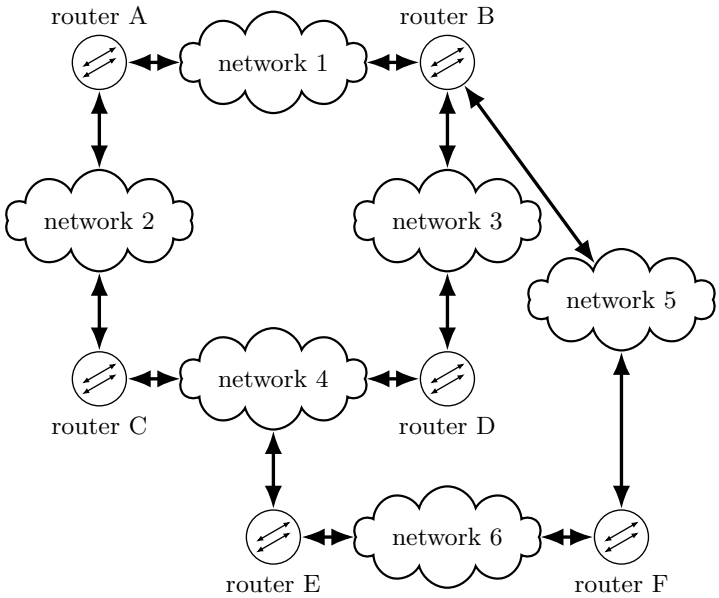
i. (5 points) Suppose the routes are setup using the distance vector protocol and then router B goes offline. After the routers that are connected to the same networks as it detect this, identify which routers might update their distances *to network 1, 2, 3, 4, 5 and/or 6* in their distance vectors as a direct or indirect result of router B going offline. **Select all that apply.**

- ■ A *longer route to network 3,5*
- ☐ C *might change next hop to network 5, but not distance*
- ■ D *longer router to network 1*
- ☐ E *might change next hop to network 1, but not distance*
- ■ F *longer route to network 1*

ii. (4 points) If router B's link and router F's link to network 5 went offline (but other network connections remained active), then the count-to-infinity problem could result in an indefinite change of distance vector updates. Assuming no measures are taken to resolve the count-to-infinity issue, which of the following might be true? **Select all that apply.**

- ■ the route from router A to network 5 could go in a loop through routers B, then D, then C, then back to A (without going through other routers) *as long as router F also disconnecte from network 5, no shorter path to discover to break this loop*
- ☐ the route from router A to network 5 could go in a loop through routers C, then E, then F, then E, then back to C (without going through other routers) *prevented by poison inverse*
- ■ the route from router C to network 5 could go in a loop through router D, then E, then back to C (without going through other routers)
- ☐ there would indefinitely be no working route from router B to network 6

(b) (8 points) (Reproduced from previous page:



router A is connected to	network 1
and	network 2
router B is connected to	network 1
and	network 3
and	network 5
router C is connected to	network 2
and	network 4
router D is connected to	network 3
and	network 4
router E is connected to	network 4
and	network 6
router F is connected to	network 5
and	network 6

)

Rather than decide routes using a distance-vector or link-state routing protocol, we could decide which connections from one router to another to enable using a spannning tree protocol.

Suppose we compare the results of a spanning-tree protocol (with some assignment of ID numbers to the routers) to using a link-state algorithm like OSPF where each router is considered to have a cost of 1 to reach the networks it is connected to and any router attached to any of those networks.

Give an example of a set of paths the OSPF-like protocol could select that the spanning tree protocol would not, even if we tweaked the spanning tree protocol by selecting different ordering for the routers (for example, to select a particular router as the root of the spanning tree).

Solution: example: B-1, B-5-F-6, A-1, A-2-C-4-E-6; any paths that result in a cycle when combined but are shortest alone

5. Suppose two hosts A and B are communicating via TCP over IPv6 through a router C.

(a) (6 points) Suppose they both have direct wired connections to the router C for which they use a maximum segment size (maximum size of data in a TCP packet) of 1000 bytes. In the largest TCP packets they transmit, they have 1000 bytes of data, plus a 40-byte IPv6 header, plus a 32-byte TCP header (including 12 bytes of TCP options).

Each wired connection has a latency of 10 microseconds, a packet with the maximum amount of data takes 1 microsecond to transmit, a packet with no data takes around 0.1 microsceonds to transmit, and the router has an average queuing latency of 5 microseconds.

If we use a fixed window size for the TCP connection (instead of normal congestion control), then approximately what should that window size be in bytes to achieve the maximum transmission rate? You may leave your answer as unsimplified arithmetic expression.

Solution: 1000 bytes / 1 microsecond * (1 + 10 + 5 + 1 + 10 + .1 + 10 +5 + .1 + 10)

(b) (4 points) Suppose instead of having direct connections to C, A and B use a VPN-style tunnel to connect to C. This tunnel is implemented using GRE over IPv6, with a 40-byte outer IPv6 header specifying a destination of C, a 4-byte GRE header, followed by IPv6 and TCP headers in the original format.

This reduces the maximum segment size to 996 bytes. What is the total increase in **IPv6 packet bytes** sent from A to B, in percent *was misspelled precent on exam*? Please leave your answer as an unsimplified arithmetic expression.

Solution: old per packet = 1000+32+40; new per packet = 996+32+40+4+40; 1000/996 new packets per old packet, so 1000/996*(996+32+40+4+40)/(1000+32+40)*100 percent

6. Suppose we have:

   - a router that:
     - is connected to a local 10.0.0.0/24 Ethernet network with IP address 10.0.0.1
     - is connected the Internet with IP address 198.51.100.45 and gateway 198.51.100.1
     - uses network address translation to allow machines on the 10.0.0.0/24 network to access the Internet using the router's Internet IP address
   - a machine with a web browser in the local Ethernet network with IP address 10.0.0.4
   - a machine with a web proxy in the local Ethernet netiwork with IP address 10.0.0.7

   The web browser on 10.0.0.4 makes the following request to the web proxy:

```
GET http://quux.foo.com/ HTTP/1.1
Host: quux.foo.com
Cookie: foo=bar
```

(a) (8 points) Suppose the web proxy looks up the IPv4 address of `quux.foo.com`. Rather than contacting a recursive DNS resolver, it makes DNS queries itself starting with a root server via IPv4. It sends three separate queries. **Complete the table** with plausible information about these queries and their response. (You may not need all lines.) *one possible answer:*
a.gtld-servers.net was miswritten a.gtld-server.net once in the original exam

destination hostname	query question	response
a.root-servers.net	quux.foo.com A	`com. IN NS a.gtld-servers.net.` `a.gtld-servers.net. IN A 192.5.6.30`
a.gtld-server.net	quux.foo.com IN A	foo.com IN NS ns.bar.com ns.bar.com. IN A 1.2.3.4
ns.bar.com	`quux.foo.com IN A`	quux.foo.com. `IN A 2.3.4.5`

(b) (4 points) The web proxy sends a DNS query to a.root-servers.net (which has IPv4 address 198.41.0.4) via UDP over IPv4. Before it is processed by routers, the DNS packet the web proxy sends for this query will have a destination IP address of 198.41.0.4. When a.root-servers.net sends its response to the query, before any processing by routers, the DNS packet it sends will have a destination address of 198.51.100.45.

If not enough information is provided to answer above, explain briefly below.

Solution:

(c) (5 points) After resolving the IP address of quux.foo.com, the web proxy uses the sockets API to send a request to it. When it does so, it is likely to _____. **Select all that apply.**

- ☐ pass the local IP address of the router (10.0.0.1) to the '`connect`' call for that socket
- ■ pass the default port number the HTTP protocol (80) to the '`connect`' call for that socket
- ☐ pass the local IP address of the router (10.0.0.1) to the '`bind`' call for that socket.
- ☐ pass the default port number the HTTP protocol (80) to the '`bind`' call for that socket
- ☐ call the '`listen`' function on that socket, so it can use it as a server socket and accept connections

(d) (4 points) (Reproduced from previous page:

- a router that:
  - is connected to a local 10.0.0.0/24 Ethernet network with IP address 10.0.0.1
  - is connected the Internet with IP address 198.51.100.45 and gateway 198.51.100.1
  - uses network address translation to allow machines on the 10.0.0.0/24 network to access the Internet using the router's Internet IP address
- a machine with a web browser in the local Ethernet network with IP address 10.0.0.4
- a machine with a web proxy in the local Ethernet netiwork with IP address 10.0.0.7

)

The web browser on 10.0.0.4 makes the following request to the web proxy:

```
GET http://quux.foo.com/ HTTP/1.1
Host: quux.foo.com
Cookie: foo=bar

```

Suppose the web proxy makes a request to the quux.foo.com server, then gets the reply:

```
HTTP/1.1 200 OK
Cache-Control: max-age=600
Content-Type: text/plain
Content-Length: 10000
E-Tag: "abcd-12345"

...
```

(where `...` represents some unshown bytes).

Which of the following might the web proxy do with this reply? **Select all that apply.**

- ■ send it to 10.0.0.4 again when it makes another request for the same URL immediately afterwards, without sending another request to `quux.foo.com`
- ■ send it to 10.0.0.4 again when it makes another request for the same URL later after receiving a "`304 Not Modified`" response from `quux.foo.com`
- ☐ send it to 10.0.0.4 again when it makes a request for the URL `http://www.quux.foo.com/` without sending another request to `quux.foo.com` or `www.quux.foo.com`
- ■ send it to another web browser (for example, on 10.0.0.3) that makes a request shortly after 10.0.0.4's request that is identical, except that it has a different value for the `Cookie` header, without sending another request to `quux.foo.com`

7. Suppose there are four nodes A, B, C, D, and E using a Wi-Fi-style wireless network to communicate. Nodes are in radio range to reach other as indicated in the following table:

	A	B	C	D	E
A	✓	✓	✓		✓
B	✓	✓		✓	✓
C	✓		✓	✓	✓
D		✓	✓	✓	✓
E	✓	✓	✓	✓	✓

C and E supposed to be symmetric in chart above, not as printed. The printed version conflicts with the information re: who's transmitting to who

Assume that

- if a node receives transmissions from two nodes at the same time, the result is a collision so neither transmission is intelligible;
- a node cannot receive other transmissions while it is also transmitting; and
- that propagation delays are very small compared to message transmission times or the length of time used for carrier sense.

(a) (4 points) Suppose we are using carrier sense with randomized exponential backoff and no ready-to-send/clear-to-send protocol or similar.

Also suppose that if A was transmitting a large stream of data to B, then the effective bandwidth of that stream is almost the capacity of the wireless channel.

If, instead:

- A transmitting a large stream of data to B, **and**
- E transmitting a large stream of data to C, **and**
- B transmitting a large stream of data to E

at the same time. Assume they always have packets queued to transmit.

The combined effective bandwidth (after accounting for retransmissions, lost packets, etc.) of these three streams will most likely be _____.

- ○ very small — almost all packets will not be transmitted successfully
- ■ much less than the total bandwidth with one sender (less than 80% of the normal bandwidth), but still significant *probably better answer*
- ○ measurably less than the total bandwidth with one sender (maybe by a few percent), but almost all the channel's capacity *also accept*
- ○ essentially the same as the total bandwidth with one sender
- ○ more than the capacity of the wireless channel (more bandwidth because of three separate streams)

(b) (4 points) For which of the following scenarios would ready-to-send/clear-to-send improve performance? **Select all that apply.**

- ■ A is transmitting a large stream of data to B while D is transmitting a large stream of data to B
- ■ A is transmitting a large stream of data to B while D is transmitting a large stream of data to C
- ☐ B is transmitting a large stream of data to A while D is transmitting a large stream of data to C
- ☐ A is transmitting two large streams of data, one to B and one to C

8. Suppose we have a channel on which we can transmit bits, and we want to use it to transmit distinct messages.

   Consider the following proposals:

   A. have messages be a fixed size and followed by a 32-bit checksum;
   B. have messages be fixed sized, with the bit pattern `01111111111111111110` added before the message contents and a 32-bit checksum added after it;
   C. have messages be variable sized, and prefixed by their length (as a 32-bit number) and a 32-bit checksum;
   D. have messages be variable sized, and prefixed by the bit pattern `0111110` and have all `011111`s in the message body replaced with `0111111`

   Identify which of the protocols above have the following properties.

(a) (2 points) Allows most of the bits for a message to start being sent before the entire contents of the message is known. **Select all that apply.**

■ protocol A ■ protocol B ☐ protocol C *possible if length known, so accepted* ■ protocol D

(b) (2 points) Allows a receiver to discard almost all messages corrupted by bits being flipped. **Select all that apply.**

■ protocol A ■ protocol B ■ protocol C ☐ protocol D

(c) (2 points) After bits are omitted or duplicated to a clock synchronization issue, allows a receiver to efficiently locate and receive the next non-corrupted message. **Select all that apply.**

☐ protocol A ■ protocol B ☐ protocol C ■ protocol D

9. (7 points) TCP congestion control based on observing packet drops is sensitive to switch and router buffer sizes.

   Increasing the buffer size on routers carrying a long TCP flows that use drop-based congestion control is likely to _____. **Select all that apply.**

   ☐ allow the long TCP flows to react faster to changes in available bandwidth

   ■ allow the long TCP flows' throughput to be less affected by bursts of short flows using the router

   ■ allow the long TCP flows' latency to be less affected by bursts of short flows using the router *accepted either answer; not clear in absolute terms, true in relative terms*

   ■ increase the average TCP window size for the long TCP flows *increased queuing latency (since drop-based congestion control tends to fill buffer), since window size is prpoprtional to bandwidth times RTT, it will increase (even though overall bandwidth does not really)*

   ■ increase the latency of UDP traffic using the same routers

   ☐ increase the number of duplicate ACKs and/or timeouts the long TCP flows observe when the available bandwidth is constant *lowering effect of cross-traffic, so most likely; but accepted selected on theory that extreme buffer sizes would trigger timeouts*

   ■ increase the round-trip time of the long TCP flows *increase queuing latency*